

THEORY OF MACHINES - II

III B. Tech. I Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME24	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course is intended to deal with the forces and their effects, while acting upon the machine parts in motion. The knowledge of this subject is very essential for an engineer in designing the various parts of a machine. Study of gyroscopes is very much helpful to learn the precession and stability of moving vehicle such as aircrafts, Naval ships and automobiles. This course helps to analyse and evaluate the force and torques that causes of motions in mechanisms. It provides knowledge on clutches, brakes and dynamometers, balancing of machines and application of vibrations.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Analyse the effect of gyroscopic couples on the stability of aeroplane, naval ship and automobiles moving in a curved path and Static and dynamic analysis of 4-link mechanism.
2. Analyse the frictional forces and torques transmitted by various components such as clutches, brakes and dynamometers.
3. Demonstrate governing systems that control the speed of engines using the inertia phenomena and centrifugal forces and analyse different types of governors.
4. Find the position and magnitude of unbalanced masses for rotating and reciprocating parts of the engine
5. Determine frequency of various machine members subjected to free and forced vibrations.

UNIT-I GYROSCOPIC COUPLE, STATIC & DYNAMIC FORCE ANALYSIS

Precessional angular motion: Gyroscopic couple – effect of precessional motion on the stability of moving vehicles such as motorcycle – motorcar – aeroplanes and ships.

Static and Dynamic Force Analysis: Static force analysis of planar mechanisms – Analytical Method – D'Alembert's principle, Dynamic Analysis of 4-link mechanism, Single Slider Crank Mechanism. Engine Force Analysis – Piston Effort, Crank Effort of reciprocating Engine.

UNIT-II FRICTION, BRAKES & DYNAMOMETERS

Friction: Pivots and collars – uniform pressure, uniform wear – friction circle and friction axis: lubricated surfaces – boundary friction – film lubrication. Clutches – Types – Single plate, multi-plate, cone and centrifugal clutches.

Brakes and Dynamometers: Types of brakes: Simple block brake, band and block brake – internal expanding shoe brake – effect of braking of a vehicle. Dynamometers – absorption and transmission types.

UNIT-III TURNING MOMENT DIAGRAM AND FLYWHEELS, GOVERNORS

Turning Moment Diagram and Flywheels: Turning moment diagram – fluctuation of energy – flywheels – energy stored and their design – flywheel in punching press.

Governors: Types of governors – Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. Sensitiveness, isochronisms and hunting – stability of governors.

UNIT-IV	BALANCING
Balancing of rotating masses – Analytical and graphical methods. Balancing of reciprocating masses – Primary, Secondary, and higher balancing. Examination of multi cylinder-in-line engines for primary and secondary balancing- locomotive balancing – hammer blow – Swaying couple – variation of tractive effort.	
UNIT-V	VIBRATIONS
Vibrations: Types of vibratory motion, types of free vibrations, natural frequency of free longitudinal and transverse vibrations – natural frequency of free transverse vibrations over a simply supported shaft and fixed shaft due to concentrated and distributed loads. Critical or whirling speed of a shaft – Natural frequency of torsional vibrations of single rotor, two rotor and three rotor systems.	
Text Books:	
1. S.S. Rattan, Theory of Machines, 4th Edition, Tata McGraw-Hill Education India Pvt. Ltd., 2017. 2. Sadhu Singh, Theory of Machines: Kinematics and Dynamics, 3rd Edition, Pearson Education India, 2012.	
Reference Books:	
1. J.S. Rao and R.V. Duggipati, Mechanism and Machine Theory, New Age International (P) Ltd., 1992. 2. John J. Uicker, Jr., Gordon R. Pennock, Joseph E. Shigley, Theory of Machines, 5th Edition, Oxford University Press, 2017. 3. R.S. Khurmi, Theory of Machines, 14th Edition, S. Chand & Company, New Delhi, India, 2018.	